

Mini Project – Customer Sales Insights

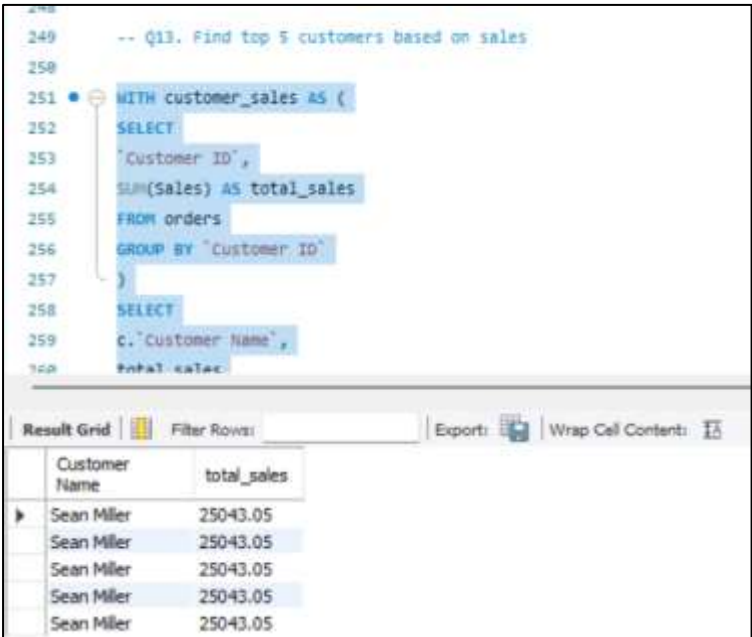
*(These queries are included in the advanced_sql_queries.sql file.
The following queries are from Q13 to Q17.)*

1. Who are the top 5 customers?

SQL Query:

```
WITH customer_sales AS (  
SELECT  
  `Customer ID`,  
  SUM(Sales) AS total_sales  
FROM orders  
GROUP BY `Customer ID`  
)  
SELECT  
  c.`Customer Name`,  
  total_sales  
FROM customer_sales cs  
JOIN customers c  
ON cs.`Customer ID` = c.`Customer ID`  
ORDER BY total_sales DESC  
LIMIT 5;
```

Output:



The screenshot shows a SQL query editor with the following text:
-- Q13. Find top 5 customers based on sales
WITH customer_sales AS (
 SELECT
 `Customer ID`,
 SUM(Sales) AS total_sales
 FROM orders
 GROUP BY `Customer ID`
)
SELECT
 c.`Customer Name`,
 total_sales
FROM customer_sales cs
JOIN customers c
ON cs.`Customer ID` = c.`Customer ID`
ORDER BY total_sales DESC
LIMIT 5;

Below the query editor is a 'Result Grid' with the following data:

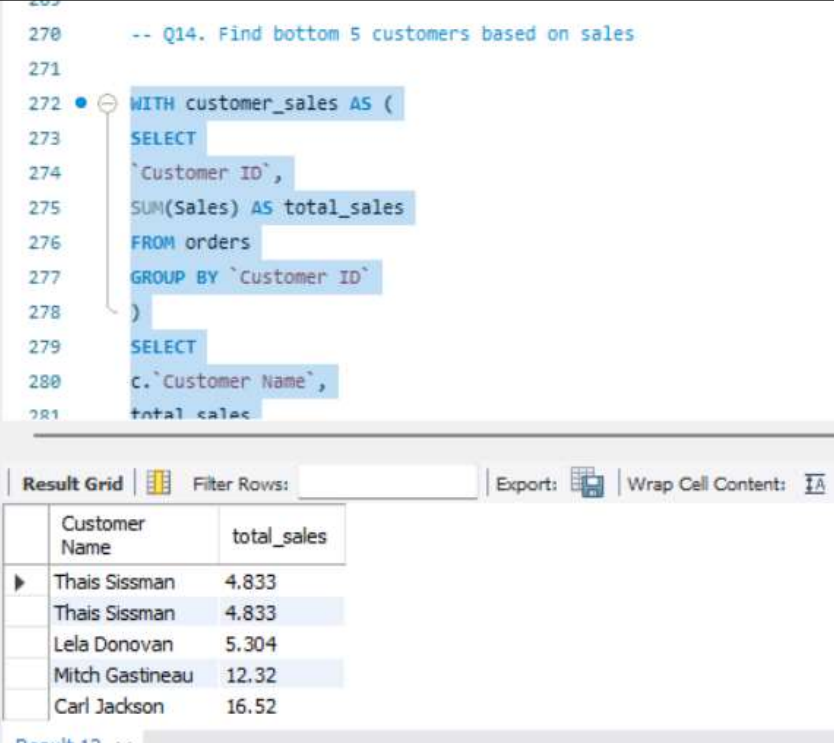
	Customer Name	total_sales
▶	Sean Miller	25043.05
	Sean Miller	25043.05
	Sean Miller	25043.05
	Sean Miller	25043.05
	Sean Miller	25043.05

2. Who are the bottom 5 customers?

SQL Query:

```
WITH customer_sales AS (  
    SELECT  
        `Customer ID`,  
        SUM(Sales) AS total_sales  
    FROM orders  
    GROUP BY `Customer ID`  
)  
SELECT  
    c.`Customer Name`,  
    total_sales  
FROM customer_sales cs  
JOIN customers c  
ON cs.`Customer ID` = c.`Customer ID`  
ORDER BY total_sales ASC  
LIMIT 5;
```

Output:



```
270 -- Q14. Find bottom 5 customers based on sales  
271  
272 WITH customer_sales AS (  
273     SELECT  
274         `Customer ID`,  
275         SUM(Sales) AS total_sales  
276     FROM orders  
277     GROUP BY `Customer ID`  
278 )  
279 SELECT  
280     c.`Customer Name`,  
281     total_sales
```

Customer Name	total_sales
Thais Sissman	4.833
Thais Sissman	4.833
Lela Donovan	5.304
Mitch Gastineau	12.32
Carl Jackson	16.52

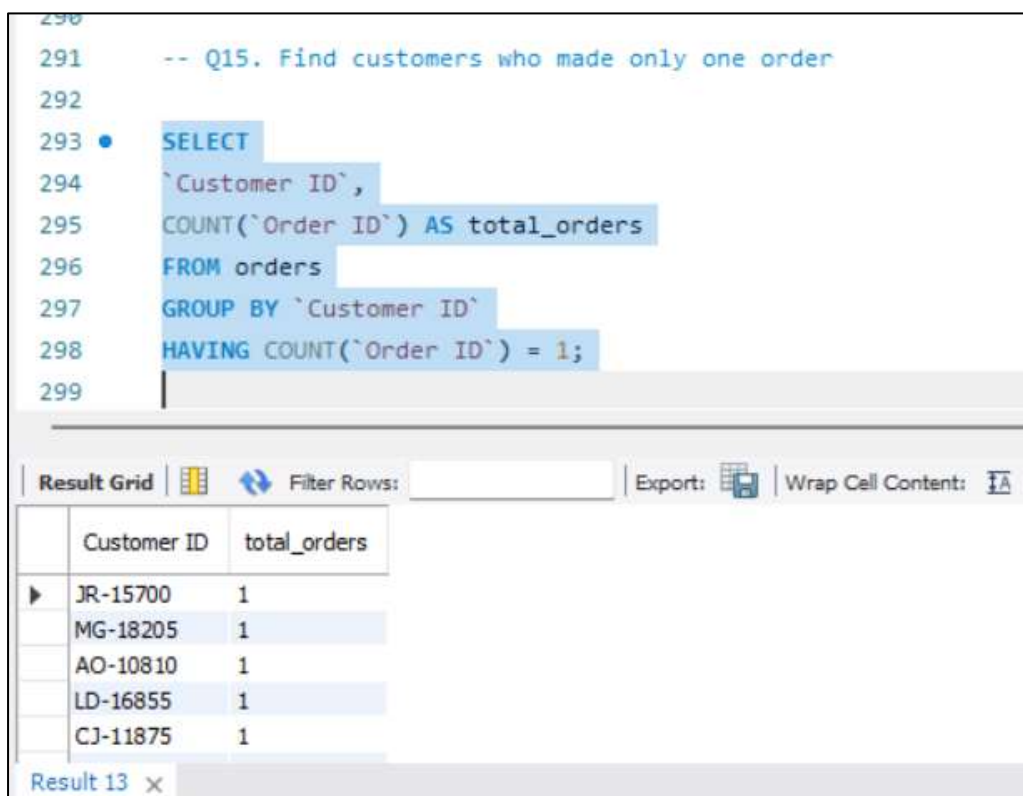
Result 12 x

3. Which customers made only one order?

SQL Query:

```
SELECT  
`Customer ID`,  
COUNT(`Order ID`) AS total_orders  
FROM orders  
GROUP BY `Customer ID`  
HAVING COUNT(`Order ID`) = 1;
```

Output:



```
290  
291 -- Q15. Find customers who made only one order  
292  
293 • SELECT  
294 `Customer ID`,  
295 COUNT(`Order ID`) AS total_orders  
296 FROM orders  
297 GROUP BY `Customer ID`  
298 HAVING COUNT(`Order ID`) = 1;  
299
```

Customer ID	total_orders
JR-15700	1
MG-18205	1
AO-10810	1
LD-16855	1
CJ-11875	1

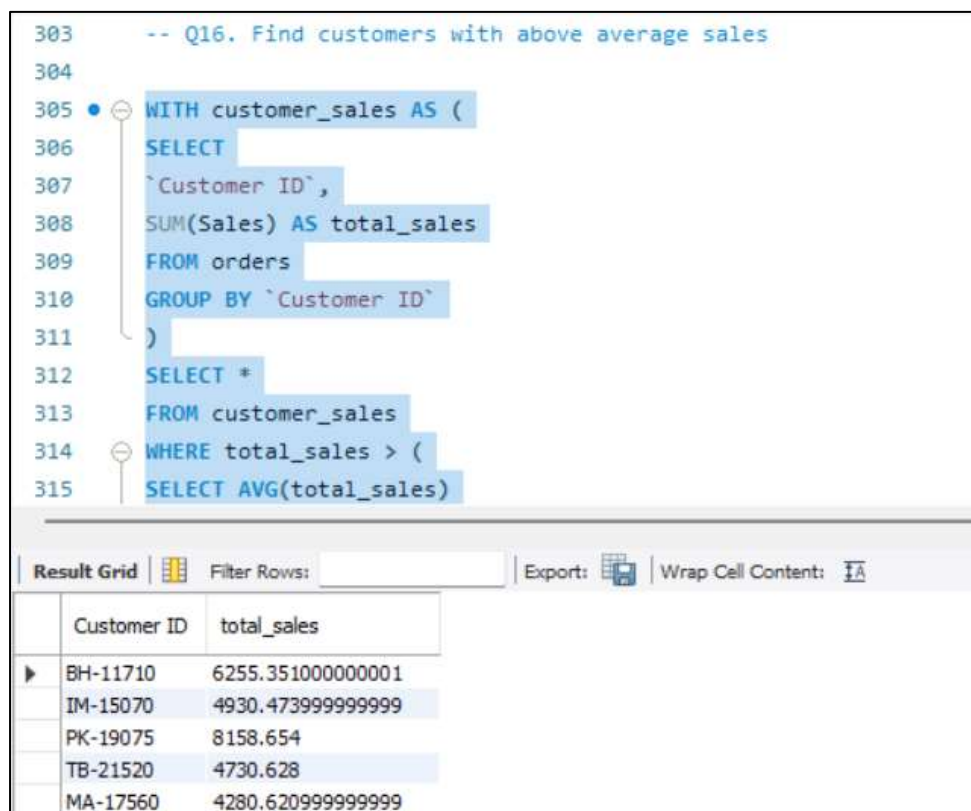
Result 13 x

4. Which customers have above-average sales?

SQL Query:

```
WITH customer_sales AS (  
  SELECT  
    `Customer ID`,  
    SUM(Sales) AS total_sales  
  FROM orders  
  GROUP BY `Customer ID`  
)  
SELECT *  
FROM customer_sales  
WHERE total_sales > (  
  SELECT AVG(total_sales)  
FROM customer_sales  
);
```

Output:



```
303 -- Q16. Find customers with above average sales  
304  
305 WITH customer_sales AS (  
306   SELECT  
307     `Customer ID`,  
308     SUM(Sales) AS total_sales  
309   FROM orders  
310   GROUP BY `Customer ID`  
311 )  
312 SELECT *  
313 FROM customer_sales  
314 WHERE total_sales > (  
315   SELECT AVG(total_sales)
```

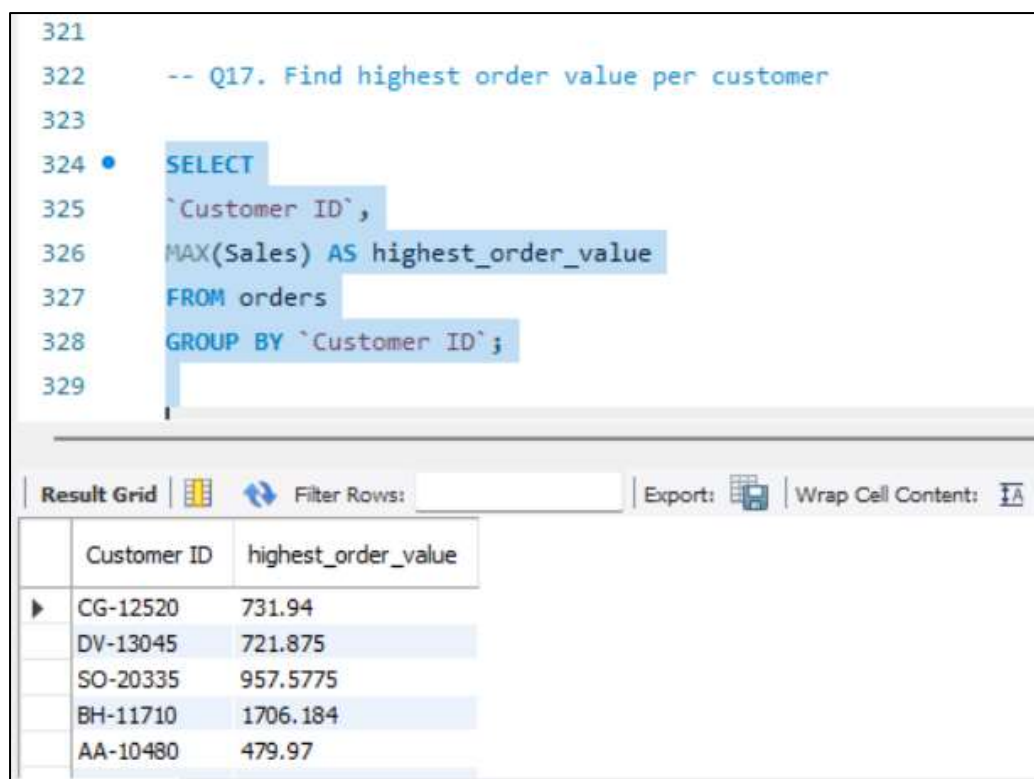
Customer ID	total_sales
BH-11710	6255.351000000001
IM-15070	4930.4739999999999
PK-19075	8158.654
TB-21520	4730.628
MA-17560	4280.6209999999999

5. What is the highest order value per customer?

SQL Query:

```
SELECT  
`Customer ID`,  
MAX(Sales) AS highest_order_value  
FROM orders  
GROUP BY `Customer ID`;
```

Output:



```
321  
322 -- Q17. Find highest order value per customer  
323  
324 • SELECT  
325 `Customer ID`,  
326 MAX(Sales) AS highest_order_value  
327 FROM orders  
328 GROUP BY `Customer ID`;  
329
```

Customer ID	highest_order_value
CG-12520	731.94
DV-13045	721.875
SO-20335	957.5775
BH-11710	1706.184
AA-10480	479.97